

SAM ZECKENDORF

sam@zeckendorf.me
(862) 220-0477

ELECTRICAL & COMPUTER ENGINEER

Experience

Apple Inc.

Lead Hardware Engineer (Homelabs)

October 2021 — Present

Cupertino, CA

- Leading a team of engineers designing the next generation of hardware for the home
 - Responsible for technical leadership and architecture of a complex system across a dozen boards and flexes facilitating actuation, proprioception, and closed-loop motion controls
 - Architected, designed, tested, and manufactured the electronics behind complex motor driver / encoder assemblies intended for consumer scale
 - Coordinated between multiple cross functional disciplines (SW/FW/Mechanical/Thermal/Controls) to deliver on complex system requirements for a novel product category
-

Space Exploration Technologies Corp.

Lead Hardware Engineer (SpaceX/Starlink)

May 2018 — October 2021

Hawthorne, CA

- Managed a team of engineers designing the next generation of hardware to connect the world to the internet via the Starlink constellation
 - Coordinated with cross functional partners across supply chain, manufacturing, test, reliability, and compliance to build an optimal domestic production line
 - Responsible for schematic capture and layout for a variety of AC/DC power supplies, WiFi routers, and phased array antennas
 - Previously, member of team responsible for designing Dragon 2 power system and conops
-

Google Inc.

Hardware Design Engineer (Nest & Daydream VR)

September 2014—January 2018

Mountain View, CA

- Designed hardware for Daydream VR wearables team from schematic capture & layout to validation and test. Responsible for 8 separate PCBAs, FPCAs, and cable assemblies within flagship product from cross functional design to factory quality and bringup.
 - Drove hardware design from marketing requirements to engineering specification, e.g. developing production test tools for measuring "motion-to-photon" latency
 - Designed several FPCs and PCBs for Nest Secure, driving schematic capture, component selection, layout and bring-up in factory/lab. Identified design/manufacturing issues at factory in OQC/IQC, and pursued through resolution to optimize yields.
 - Created and maintained MATLAB/SPICE simulations for sensitive circuitry and power modeling to allow data-driven hardware design, e.g. RX for active IR intrusion detector
 - Wrote and maintained python libraries for interfacing with various internal test equipment over USB/GPIB, governed by web app
-

Center for Engineering Education Outreach

STOMP Lego Engineering Fellow

September 2011 — December 2013

Somerville, MA

- Part of nonprofit that travels to classrooms in Cambridge and Boston Massachusetts to teach engineering fundamentals to middle and elementary school students
 - Designed lessons around important and esoteric ideas such as collaborative design, limited materials, abstract problem solving
 - See More: <http://www.legoengineering.com/about/>
-

Education

Tufts University
Bachelor of Science in Electrical Engineering

September 2010 - May 2014
Medford, MA

GPA 3.8

Selected Coursework: Feedback-Control Systems, Communication Systems, Microprocessor Architecture & Assembly Code, Digital Logic Circuits, Analog Design I & II, Physics of Solar Cells, Data Structures, Usability Engineering, Linear Algebra, Discrete Mathematics, Multivariable Calculus, Differential Equations, Music Applications on the iPad, Game Design

Projects

imPact

Co-created iPhone music application "imPact: Remix", funded by Steinway Music

Squid

Xbox game, awarded Indie Gem on Joystiq.com (affiliate of Engadget)

Smart Hydroponics

Intelligent, learning, home-hydroponics system for city dwellers to grow fresh produce. White paper available

Languages & Skills

Cadence (Concept + Allegro)
Altium

C/C++/C#/Objective-C
Matlab

Python
VHDL